

SOLIDWORKS API (NET VBA C#) — Reference

(137/337): IIndentFeatureData-Interference

SOLIDWORKS API Documentation (converted from CHM)

IIndentFeatureData (Type)

IIndentFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IIndentFeatureData Interface

Allows access to an indent feature.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IIndentFeatureData
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData
```

C#

```
public interface IIndentFeatureData
```

C++/CLI

```
public interface class IIndentFeatureData
```

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData](#).

Example

[Insert Indent Feature \(C#\)](#)
[Insert Indent Feature \(VB.NET\)](#)
[Insert Indent Feature \(VBA\)](#)

Accessors

[IFeature::GetDefinition](#)

Access Diagram

[IndentFeatureData](#)

See Also

[IIndentFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertIndent Method \(\)](#)

IIndentFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IIndentFeatureData Interface

For a list of all members of this type, see [IIndentFeatureData members](#).

Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this indent feature.
 GetBodiesCount	Gets the number of solid or surface bodies for the tool body region .

 ReleaseSelectionAccess	Releases access to the selections for this indent feature.
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See Also

[IIndentFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertIndent Method \(\)](#)

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IIndentFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IIndentFeatureData Interface

The following tables list the members exposed by [IIndentFeatureData](#).

Public Properties

Name	Description
 Clearance	Gets or sets the clearance between the target and tool bodies in this indent feature.
 ClearanceDirection	Gets or sets the direction of the clearance for this indent feature.
 CutDirection	Gets or sets whether to flip the side of the cut for this indent feature.
 IsCut	Gets or sets whether to remove the intersection area of the target body .
 SelectionState	Gets or sets the side of the model to keep or remove.
 TargetBody	Gets or sets the solid or surface body to indent.
 Thickness	Gets or sets the thickness of the indent feature.
 ToolBodyRegion	Gets or sets the tool body region for the indent feature.

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Public Methods

Name	Description
 AccessSelections	Gains access to the selections that define this indent feature.
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 ReleaseSelectionAccess	Releases access to the selections for this indent feature.

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See Also

[IIndentFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertIndent Method \(\)](#)

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IIndentFeatureData Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IIndentFeatureData Interface

For a list of all members of this type, see [IIndentFeatureData members](#).

Public Properties

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SelectionState	Gets or sets the side of the model to keep or remove.

[TargetBody](#)

Gets or sets the solid or surface body to indent.

[Thickness](#)

Gets or sets the thickness of the indent feature.

[ToolBodyRegion](#)

Gets or sets the tool body region for the indent feature.

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See Also

[IIndentFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IFeatureManager::InsertIndent Method \(\)](#)

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AccessSelections Method (IIndentFeatureData)

[See Also](#)
[Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : AccessSelections Method (IIndentFeatureData)

Gains access to the selections that define this indent feature.

.NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections(
    ByVal TopDoc As System.Object, _
    ByVal Component As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData
Dim TopDoc As System.Object
Dim Component As System.Object
Dim value As System.Boolean
```

```
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(
    System.object TopDoc,
    System.object Component
)
```

C++/CLI

```
System.bool AccessSelections(
    & System.Object^ TopDoc,
    & System.Object^ Component
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*TopDoc*Top-level document (see **Remarks**)*Component*Component in which to modify the feature (see **Remarks**)

Return Value

True if the selections are successfully accessed, false if not

Visual Basic for Applications (VBA) Syntax

See [IIndentFeatureData::AccessSelections](#).

Example

[Insert Indent Feature \(C#\)](#)
[Insert Indent Feature \(VB.NET\)](#)
[Insert Indent Feature \(VBA\)](#)

Remarks

To modify a feature in a...	Then...
Part	- TopDoc argument is the ModelDoc2 object for the part - Component argument is Nothing or null
Assembly	- TopDoc is the ModelDoc2 object for the assembly - Component argument is the Component2 object in which to modify the feature

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [IIndentFeatureData::ReleaseSelectionAccess](#) if you did not

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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Clearance Property (IIndentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : Clearance Property (IIndentFeatureData)

Gets or sets the clearance between the target and tool bodies in this indent feature.

.NET Syntax

Visual Basic (Declaration)

Property Clearance As System.Double

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Double
```

```
instance.Clearance = value
```

```
value = instance.Clearance
```

C#

```
System.double Clearance {get; set;}
```

C++/CLI

```
property System.double Clearance {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Clearance between the target and tool bodies

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::Clearance](#).

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)
[IIndentFeatureData::ClearanceDirection Property \(\)](#)
[IIndentFeatureData::TargetBody Property \(\)](#)
[IIndentFeatureData::ToolBodyRegion Property \(\)](#)

Availability

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ClearanceDirection Property (IIndentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : ClearanceDirection Property (IIndentFeatureData)

Gets or sets the direction of the [clearance](#) for this indent feature.

.NET Syntax

Visual Basic (Declaration)

Property ClearanceDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Boolean
```

```
instance.ClearanceDirection = value
```

```
value = instance.ClearanceDirection
```

C#

```
System.bool ClearanceDirection {get; set;}
```

C++/CLI

```
property System.bool ClearanceDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to reverse the direction of clearance, false to not

Visual Basic for Applications (VBA) Syntax

See [IIndentFeatureData::ClearanceDirection](#).

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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CutDirection Property (IIndentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : CutDirection Property (IIndentFeatureData)

Gets or sets whether to flip the side of the cut for this indent feature.

.NET Syntax

Visual Basic (Declaration)

```
Property CutDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Boolean
```

```
instance.CutDirection = value
```

```
value = instance.CutDirection
```

C#

```
System.bool CutDirection {get; set;}
```

C++/CLI

```
property System.bool CutDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to flip the side of the cut, false to not

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::CutDirection](#).

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)
[IIndentFeatureData::IsCut Property \(\)](#)

Availability

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GetBodiesCount Method (IIndentFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : GetBodiesCount Method (IIndentFeatureData)

Gets the number of solid or surface [bodies](#) for the [tool body region](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetBodiesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Integer
```

```
value = instance.GetBodiesCount()
```

C#

```
System.int GetBodiesCount()
```

C++/CLI

```
System.int GetBodiesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of solid or surface bodies for the tool body region

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::GetBodiesCount](#).

Example

[Insert Indent Feature \(C#\)](#)
[Insert Indent Feature \(VB.NET\)](#)
[Insert Indent Feature \(VBA\)](#)

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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IsCut Property (IIndentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : IsCut Property (IIndentFeatureData)

Gets or sets whether to remove the intersection area of the [target body](#).

.NET Syntax

Visual Basic (Declaration)

```
Property IsCut As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Boolean
```

```
instance.IsCut = value
```

```
value = instance.IsCut
```

C#

```
System.bool IsCut {get; set;}
```

C++/CLI

```
property System.bool IsCut {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to remove the intersection area of the target body, false to not

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::IsCut](#).

Remarks

If this property is set to true, then there is no [thickness](#), but [clearance](#) is applied.

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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ReleaseSelectionAccess Method (IIndentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : ReleaseSelectionAccess Method (IIndentFeatureData)

Releases access to the selections for this indent feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::ReleaseSelectionAccess](#).

Remarks

[IIndentFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define this feature. Use this method to restore the rollback state if you did not modify the feature or use [IFeature::ModifyDefinition](#) if you did.

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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SelectionMode Property (IIndentFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : SelectionState Property (IIndentFeatureData)

Gets or sets the side of the model to keep or remove.

.NET Syntax

Visual Basic (Declaration)

Property SelectionState As System.Integer

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Integer
```

```
instance.SelectionState = value
```

```
value = instance.SelectionState
```

C#

```
System.int SelectionState {get; set;}
```

C++/CLI

```
property System.int SelectionState {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Side of the model to keep or remove as defined in [swIndentSelectionState_e](#)

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::SelectionState](#).

Remarks

Setting this property inverts the side of the [target body](#) to indent.

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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TargetBody Property (IIndentFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : TargetBody Property (IIndentFeatureData)

Gets or sets the solid or surface body to indent.

.NET Syntax

Visual Basic (Declaration)

Property TargetBody As System.Object

```
Dim instance As IIndentFeatureData
Dim value As System.Object

instance.TargetBody = value

value = instance.TargetBody
```

C#

```
System.Object TargetBody {get; set;}
```

C++/CLI

```
property System.Object^ TargetBody {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Solid or surface [body](#) to indent

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::TargetBody](#).

Example

[Insert Indent Feature \(C#\)](#)
[Insert Indent Feature \(VB.NET\)](#)
[Insert Indent Feature \(VBA\)](#)

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)
[IIndentFeatureData::IsCut Property \(\)](#)
[IIndentFeatureData::SelectionMode Property \(\)](#)
[IIndentFeatureData::ToolBodyRegion Property \(\)](#)

Availability

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Thickness Property (IIndentFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : Thickness Property (IIndentFeatureData)

Gets or sets the thickness of the indent feature.

.NET Syntax

Visual Basic (Declaration)

```
Property Thickness As System.Double
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IIndentFeatureData
Dim value As System.Double

instance.Thickness = value

value = instance.Thickness
```

C#

```
System.double Thickness {get; set;}
```

C++/CLI

```
property System.double Thickness {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Thickness

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::Thickness](#).

Example

[Insert Indent Feature \(C#\)](#)
[Insert Indent Feature \(VB.NET\)](#)
[Insert Indent Feature \(VBA\)](#)

Remarks

This property applies to solid bodies only.

If [IIndentFeatureData::IsCut](#) is set to true (i.e., remove the intersection area of the target body), then there is no thickness.

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)

Availability

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ToolBodyRegion Property (IIndentFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IIndentFeatureData Interface](#) : ToolBodyRegion Property (IIndentFeatureData)

Gets or sets the tool body region for the indent feature.

.NET Syntax

Visual Basic (Declaration)

```
Property ToolBodyRegion As System.Object
```

Visual Basic (Usage)



```
Dim instance As IIndentFeatureData  
Dim value As System.Object
```

```
instance.ToolBodyRegion = value
```

```
value = instance.ToolBodyRegion
```

C#

```
System.object ToolBodyRegion {get; set;}
```

C++/CLI

```
property System.Object^ ToolBodyRegion {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Tool body region, which can be an array of [faces](#) or [vertices](#)

Visual Basic for Applications (VBA) Syntax

See [IndentFeatureData::ToolBodyRegion](#).

Example

[Insert Indent Feature \(C#\)](#)
[Insert Indent Feature \(VB.NET\)](#)
[Insert Indent Feature \(VBA\)](#)

See Also

[IIndentFeatureData Interface](#)
[IIndentFeatureData Members](#)
[IIndentFeatureData::GetBodiesCount Method \(\)](#)
[IIndentFeatureData::TargetBody Property \(\)](#)

Availability

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IInstanceToVaryOptions (Type)

IInstanceToVaryOptions Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IInstanceToVaryOptions Interface

Allows access to options for varying the dimensions and locations of instances in patterns for parts.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IInstanceToVaryOptions

Visual Basic (Usage)

Dim instance As [IInstanceToVaryOptions](#)

C#

public interface IInstanceToVaryOptions

C++/CLI

public interface class IInstanceToVaryOptions

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions](#).

Example

```
'VBA
'Preconditions:
' 1. Open public_documents\samples\tutorial\apilbox.sldprt.
' 2. Open the Immediate window.
'Postconditions:
' 1. Creates a bi-directional linear pattern of a CBORE feature.
' 2. Specifies to vary pattern instances.
' 3. Sets the spacing increment of instances in both directions.
' 4. Modifies the spacing of instance #3.
' 5. Increments a near countersink diameter for all pattern instances in Direction 1.
' 6. Modifies the near countersink diameter for instance #3.
' 7. Inspect the Immediate window and the graphics area.
' NOTE: Because the part is used elsewhere, do not save changes.
'=====

Option Explicit

Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swSelMgr As SldWorks.SelectionMgr
Dim feat As SldWorks.Feature
Dim LPFeatData As SldWorks.LinearPatternFeatureData
Dim boolstatus As Boolean
Dim swInstOpts As SldWorks.InstanceToVaryOptions

Sub main()

    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swSelMgr = swModel.SelectionManager

    boolstatus = swModel.Extension.SelectByRay(4.39252690830472E-02, 2.99751045230323E-02, 1.26537412306789E-02,
-0.400036026779312, -0.515038074910024, -0.758094294050284, 6.33983676613175E-04, 1, True, 1, 0)
    boolstatus = swModel.Extension.SelectByRay(-1.78361123644777E-02, 2.98747083165836E-02, -3.88627220451099E-02,
-0.400036026779312, -0.515038074910024, -0.758094294050284, 6.33983676613175E-04, 1, True, 2, 0)
    boolstatus = swModel.Extension.SelectById2("CBORE for #6 Binding Head Machine Screw1", "BODYFEATURE", 0, 0, 0, True, 4, Nothing, 0,
```

```

swModel.ActivateSelectedFeature
swModel.ClearSelection2 True
boolstatus = swModel.Extension.SelectByID2("CBORE for #6 Binding Head Machine Screw1", "BODYFEATURE", 0, 0, 0, False, 4, Nothing, (
boolstatus = swModel.Extension.SelectByRay(4.39252690830472E-02, 2.99751045230323E-02, 1.26537412306789E-02,
-0.400036026779312, -0.515038074910024, -0.758094294050284, 6.33983676613175E-04, 1, True, 1, 0)
boolstatus = swModel.Extension.SelectByRay(-1.78361123644777E-02, 2.98747083165836E-02, -3.88627220451099E-02,
-0.400036026779312, -0.515038074910024, -0.758094294050284, 6.33983676613175E-04, 1, True, 2, 0)

```

```

Dim swFeat As Object
Dim swFeatMgr As Object

```

```

Set swFeatMgr = swModel.FeatureManager
Dim swFeatData As Object
Set swFeatData = swFeatMgr.CreateDefinition(swFeatureNameID_e.swFmLPattern)
swFeatData.D1EndCondition = 0
swFeatData.D1ReverseDirection = False
swFeatData.D1Spacing = 0.02
swFeatData.D1TotalInstances = 2
swFeatData.D2EndCondition = 0
swFeatData.D2PatternSeedOnly = False
swFeatData.D2ReverseDirection = False
swFeatData.D2Spacing = 0.01
swFeatData.D2TotalInstances = 2
swFeatData.GeometryPattern = False
swFeatData.VarySketch = False

```

```

Set swFeat = swFeatMgr.CreateFeature(swFeatData)

```

```

boolstatus = swModel.Extension.SelectByID2("LPattern1", "BODYFEATURE", 0, 0, 0, False, 0, Nothing, 0)
Set feat = swSelMgr.GetSelectedObject6(1, -1)
Set LPFeatData = feat.GetDefinition

```

```

'Turn on Instances To Vary
boolstatus = LPFeatData.AccessSelections(swModel, Nothing)

```

```

LPFeatData.InstancesToVary = True
Debug.Print "Vary instances?" & LPFeatData.InstancesToVary

```

```

boolstatus = feat.ModifyDefinition(LPFeatData, swModel, Nothing)

```

```

'Get and set InstanceToVaryOptions
boolstatus = LPFeatData.AccessSelections(swModel, Nothing)

```

```

Set swInstOpts = LPFeatData.GetInstanceToVaryOptions()

```

```

'Set D1 and D2 spacing increments of all pattern instances
swInstOpts.D1SpacingIncrement = 0.05
swInstOpts.D2SpacingIncrement = 0.05
Debug.Print "D1 spacing increment is " & swInstOpts.D1SpacingIncrement
Debug.Print "D2 spacing increment is " & swInstOpts.D2SpacingIncrement

```

```

'Get current D1 and D2 modified spacing value for patten instance #3 (2,2)
Debug.Print "Current D1 modified spacing value for instance #3 is " & swInstOpts.GetD1ModifiedSpacingValue(3)
Debug.Print "Current D2 modified spacing value for instance #3 is " & swInstOpts.GetD2ModifiedSpacingValue(3)

```

```

'Modify the D1 and D2 spacing value for pattern instance #3 (2,2)
boolstatus = swInstOpts.ModifyD1SpacingValue(3, swInstOpts.GetD1ModifiedSpacingValue(3) + 0.02)
boolstatus = swInstOpts.ModifyD2SpacingValue(3, swInstOpts.GetD2ModifiedSpacingValue(3) + 0.02)

```

```

'Get the new D1 and D2 modified spacing value for pattern instance #3 (2,2)
Debug.Print "New D1 modified spacing value for instance #3 is " & swInstOpts.GetD1ModifiedSpacingValue(3)
Debug.Print "New D2 modified spacing value for instance #3 is " & swInstOpts.GetD2ModifiedSpacingValue(3)

```

```

'Increment a dimension for all pattern instances in Direction 1
Dim vIncr1Dims(0) As String
Dim vIncr1DimValues(0) As Double
vIncr1Dims(0) = "Near C'Sink Dia.@Sketch2"
vIncr1DimValues(0) = 0.005

```

```

swInstOpts.SetD1IncrementDimensions vIncr1Dims, vIncr1DimValues
swInstOpts.GetD1IncrementDimensions vIncr1Dims, vIncr1DimValues

```

```

'Modify a specific dimension just for pattern instance #3 (2,2)
Dim vDims(0) As String
Dim vDimValues(0) As Double
vDims(0) = "Near C'Sink Dia.@Sketch2"
vDimValues(0) = 0.02
boolstatus = swInstOpts.ModifyDimensions(3, vDims, vDimValues)

```

```

swInstOpts.GetModifiedDimensions 3, vDims, vDimValues

```

```

'Modify the pattern feature with the instance to vary options

```

Call LPFeatData.**SetInstanceToVaryOptions**(swInstOpts)
boolstatus = feat.**ModifyDefinition**(LPFeatData, swModel, Nothing)

End Sub

 Example

[Vary Linear Pattern Instances \(VB.NET\)](#)
[Vary Linear Pattern Instances \(C#\)](#)

 Remarks

This interface is valid only if [ILinearPatternFeatureData::InstancesToVary](#) or [ICircularPatternFeatureData::InstancesToVary](#) is true.

For more information about the Instance to Vary option, see the **SOLIDWORKS Help > Parts and Features > Features > Patterns and Mirroring > Control and Modify Patterns > Varying Dimension Patterns** topic.

 Accessors

[ICircularPatternFeatureData::GetInstanceToVaryOptions](#)
[ILinearPatternFeatureData::GetInstanceToVaryOptions](#)

 Access Diagram

[InstanceToVaryOptions](#)

 See Also

[IInstanceToVaryOptions Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

IInstanceToVaryOptions Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IInstanceToVaryOptions Interface

For a list of all members of this type, see [IInstanceToVaryOptions members](#).

 Public Methods

Name	Description
 GetD1IncrementDimensions	Gets the dimensions to increment in Direction 1.
 GetD1ModifiedSpacingValue	Gets the spacing in Direction 1 of the specified modified pattern instance.
 GetD2IncrementDimensions	Gets the dimensions to increment in Direction 2.
 GetD2ModifiedSpacingValue	Gets the modified spacing in Direction 2 of the specified pattern instance.
 GetModifiedDimensions	Gets the modified dimensions of the specified pattern instance.
 GetModifiedInstances	Gets instance numbers of modified pattern instances of the specified type.
 ModifyD1SpacingValue	Modifies the spacing in Direction 1 of the specified pattern instance.
 ModifyD2SpacingValue	Modifies the spacing in Direction 2 of the specified pattern instance.
 ModifyDimensions	Modifies the specified dimensions of the specified pattern instance.
 SetD1IncrementDimensions	Sets the dimensions to increment in Direction 1.
 SetD2IncrementDimensions	Sets the dimensions to increment in Direction 2.

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 See Also

[IInstanceToVaryOptions Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

InstanceToVaryOptions Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IInstanceToVaryOptions Interface

The following tables list the members exposed by [IInstanceToVaryOptions](#).

Public Properties

Name	Description
 D1SpacingIncrement	Gets or sets the spacing increment of all pattern instances in Direction 1.
 D2SpacingIncrement	Gets or sets the spacing increment of all pattern instances in Direction 2.

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Public Methods

Name	Description
 GetD1IncrementDimensions	Gets the dimensions to increment in Direction 1.
 GetD1ModifiedSpacingValue	Gets the spacing in Direction 1 of the specified modified pattern instance.
 GetD2IncrementDimensions	Gets the dimensions to increment in Direction 2.
 GetD2ModifiedSpacingValue	Gets the modified spacing in Direction 2 of the specified pattern instance.
 GetModifiedDimensions	Gets the modified dimensions of the specified pattern instance.
 GetModifiedInstances	Gets instance numbers of modified pattern instances of the specified type.
 ModifyD1SpacingValue	Modifies the spacing in Direction 1 of the specified pattern instance.
 ModifyD2SpacingValue	Modifies the spacing in Direction 2 of the specified pattern instance.
 ModifyDimensions	Modifies the specified dimensions of the specified pattern instance.
 SetD1IncrementDimensions	Sets the dimensions to increment in Direction 1.
 SetD2IncrementDimensions	Sets the dimensions to increment in Direction 2.

[Top](#)

See Also

[IInstanceToVaryOptions Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

InstanceToVaryOptions Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IInstanceToVaryOptions Interface

For a list of all members of this type, see [IInstanceToVaryOptions members](#).

Public Properties

Name	Description
 D1SpacingIncrement	Gets or sets the spacing increment of all pattern instances in Direction 1.
 D2SpacingIncrement	Gets or sets the spacing increment of all pattern instances in Direction 2.

[Top](#)

See Also

[IInstanceToVaryOptions Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

D1SpacingIncrement Property (IInstanceToVaryOptions)

[See Also](#)



Gets or sets the spacing increment of all pattern instances in Direction 1.

.NET Syntax

Visual Basic (Declaration)

Property D1SpacingIncrement As System.Double

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions
Dim value As System.Double

instance.D1SpacingIncrement = value

value = instance.D1SpacingIncrement
```

C#

```
System.double D1SpacingIncrement {get; set;}
```

C++/CLI

```
property System.double D1SpacingIncrement {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Spacing increment in Direction 1

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::D1SpacingIncrement](#).

Example

See the [IInstanceToVaryOptions](#) examples.

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)

Availability

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D2SpacingIncrement Property (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : D2SpacingIncrement Property (IInstanceToVaryOptions)

Gets or sets the spacing increment of all pattern instances in Direction 2.

.NET Syntax

Visual Basic (Declaration)

Property D2SpacingIncrement As System.Double

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions
Dim value As System.Double

instance.D2SpacingIncrement = value
value = instance.D2SpacingIncrement
```

C#

```
System.double D2SpacingIncrement {get; set;}
```

C++/CLI

```
property System.double D2SpacingIncrement {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Spacing increment in Direction 2

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::D2SpacingIncrement](#).

Example

See the [IInstanceToVaryOptions](#) examples.

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)

Availability

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GetD1IncrementDimensions Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : GetD1IncrementDimensions Method (IInstanceToVaryOptions)

Gets the dimensions to increment in Direction 1.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetD1IncrementDimensions( _
    ByRef IncrementDimensions As System.Object, _
    ByRef IncrementDimValues As System.Object _
)
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions
Dim IncrementDimensions As System.Object
Dim IncrementDimValues As System.Object

instance.GetD1IncrementDimensions(IncrementDimensions, IncrementDimValues)
```

C#

```
void GetD1IncrementDimensions(
    out System.object IncrementDimensions,
    out System.object IncrementDimValues
)
```

C++/CLI

```
void GetD1IncrementDimensions(
    & [Out] System.Object^ IncrementDimensions,
    & [Out] System.Object^ IncrementDimValues
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncrementDimensions

Array of dimensions to increment

IncrementDimValues

Array of dimension increments in Direction 1

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::GetD1IncrementDimensions](#).

Example

See the [InstanceToVaryOptions](#) examples.

See Also

[IInstanceToVaryOptions Interface](#)

[IInstanceToVaryOptions Members](#)

Availability

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SOLIDWORKS API Help

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GetD1ModifiedSpacingValue Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : GetD1ModifiedSpacingValue Method (IInstanceToVaryOptions)

Gets the spacing in Direction 1 of the specified modified pattern instance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetD1ModifiedSpacingValue( _
    ByVal Instance As System.Integer _
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IInstanceToVaryOptions
Dim Instance As System.Integer
Dim value As System.Double
```

```
value = instance.GetD1ModifiedSpacingValue(Instance)
```

C#

```
System.double GetD1ModifiedSpacingValue(
    System.int Instance
)
```

C++/CLI



```
System.double GetD1ModifiedSpacingValue(  
& System.int Instance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

Return Value

Spacing value in Direction 1

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::GetD1ModifiedSpacingValue](#).

Example

See the [InstanceToVaryOptions](#) examples.

Remarks

To specify Instance, you can use [InstanceToVaryOptions::GetModifiedInstances](#) to learn which pattern instances are modified.

To calculate Instance:

1. Ensure that the D2PatternSeedOnly property in the pattern feature data object is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (j - 1) + (i - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- *I* = pattern instance number
- *n2* = number of instances in Direction 2
- *i* = index for Direction 1
- *j* = index for Direction 2

In the pattern's PropertyManager, find *n2* in the **Direction 2 > Spacing and Instances > Number of instances** field and find [*i,j*] in the **Modified Instances** section.

See Also

[InstanceToVaryOptions Interface](#)
[InstanceToVaryOptions Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

GetD2IncrementDimensions Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [InstanceToVaryOptions Interface](#) : GetD2IncrementDimensions Method (IInstanceToVaryOptions)

Gets the dimensions to increment in Direction 2.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetD2IncrementDimensions( _  
    ByRef IncrementDimensions As System.Object, _  
    ByRef IncrementDimValues As System.Object _  
)
```

Visual Basic (Usage)

```
Dim instance As IInstanceToVaryOptions  
Dim IncrementDimensions As System.Object  
Dim IncrementDimValues As System.Object  
  
instance.GetD2IncrementDimensions(IncrementDimensions, IncrementDimValues)
```

C#

```
void GetD2IncrementDimensions(  
    out System.object IncrementDimensions,  
    out System.object IncrementDimValues  
)
```

C++/CLI

```
void GetD2IncrementDimensions(  
& [Out] System.Object^ IncrementDimensions,  
& [Out] System.Object^ IncrementDimValues  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncrementDimensions

Array of dimensions to increment

IncrementDimValues

Array of dimension increments in Direction 2

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::GetD2IncrementDimensions](#).

See Also

[IInstanceToVaryOptions Interface](#)

[IInstanceToVaryOptions Members](#)

Availability

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SOLIDWORKS API Help

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GetD2ModifiedSpacingValue Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : GetD2ModifiedSpacingValue Method (IInstanceToVaryOptions)

Gets the modified spacing in Direction 2 of the specified pattern instance.

.NET Syntax

Visual Basic (Declaration)

```
Function GetD2ModifiedSpacingValue( _  
    ByVal Instance As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IInstanceToVaryOptions  
Dim Instance As System.Integer  
Dim value As System.Double  
  
value = instance.GetD2ModifiedSpacingValue(Instance)
```

C#

```
System.double GetD2ModifiedSpacingValue(  
    System.int Instance  
)
```

C++/CLI

```
System.double GetD2ModifiedSpacingValue(  
& System.int Instance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

Return Value

Spacing value in Direction 2

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::GetD2ModifiedSpacingValue](#).

Example

See the [InstanceToVaryOptions](#) examples.

Remarks

To specify Instance, you can use [InstanceToVaryOptions::GetModifiedInstances](#) to learn which pattern instances are modified.

To calculate Instance:

1. Ensure that the D2PatternSeedOnly property in the pattern feature data object is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (i - 1) + (j - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- *I* = pattern instance number
- *n2* = number of instances in Direction 2
- *i* = index for Direction 1
- *j* = index for Direction 2

In the pattern's PropertyManager, find *n2* in the **Direction 2 > Spacing and Instances > Number of instances** field and find [*i*,*j*] in the **Modified Instances** section.

See Also

[InstanceToVaryOptions Interface](#)
[InstanceToVaryOptions Members](#)

Availability

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GetModifiedDimensions Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : GetModifiedDimensions Method (IInstanceToVaryOptions)

Gets the modified dimensions of the specified pattern instance.

.NET Syntax

Visual Basic (Declaration)

```

Sub GetModifiedDimensions( _
    ByVal Instance As System.Integer, _
    ByRef Dimensions As System.Object, _
    ByRef Values As System.Object _
)

```

Visual Basic (Usage)



```

Dim instance As IInstanceToVaryOptions
Dim Instance As System.Integer
Dim Dimensions As System.Object
Dim Values As System.Object

instance.GetModifiedDimensions(Instance, Dimensions, Values)

```

C#

```

void GetModifiedDimensions(
    System.int Instance,
    out System.object Dimensions,
    out System.object Values
)

```

C++/CLI

```

void GetModifiedDimensions(
& System.int Instance,
& [Out] System.Object^ Dimensions,
& [Out] System.Object^ Values
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance
Pattern instance number (see **Remarks**)

Dimensions
Array of modified dimensions

Values
Array of dimension values

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::GetModifiedDimensions](#).

Example

See the [IInstanceToVaryOptions](#) examples.

Remarks

To specify Instance, you can use [IInstanceToVaryOptions::GetModifiedInstances](#) to learn which pattern instances are modified.

To calculate Instance:

1. Ensure that the D2PatternSeedOnly property in the pattern feature data object is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (j - 1) + (i - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- *I* = pattern instance number
- *n2* = number of instances in Direction 2
- *i* = index for Direction 1
- *j* = index for Direction 2

In the pattern's PropertyManager, find *n2* in the **Direction 2 > Spacing and Instances > Number of instances** field and find [*i,j*] in the **Modified Instances** section.

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)

Availability

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GetModifiedInstances Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : GetModifiedInstances Method (IInstanceToVaryOptions)

Gets instance numbers of modified pattern instances of the specified type.

.NET Syntax

Visual Basic (Declaration)

```
Function GetModifiedInstances( _  
    ByVal ModificationType As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions  
Dim ModificationType As System.Integer  
Dim value As System.Object
```

```
value = instance.GetModifiedInstances(ModificationType)
```

C#

```
System.object GetModifiedInstances(  
    System.int ModificationType  
)
```

C++/CLI

```
System.Object^ GetModifiedInstances(  
& System.int ModificationType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ModificationType
Modification type as defined by [swInstanceToVaryModificationType_e](#)

Return Value

Array of pattern instance numbers

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::GetModifiedInstances](#).

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)
[IInstanceToVaryOptions::GetModifiedDimensions Method \(\)](#)
[IInstanceToVaryOptions::GetD1ModifiedSpacingValue Method \(\)](#)
[IInstanceToVaryOptions::GetD2ModifiedSpacingValue Method \(\)](#)

Availability

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ModifyD1SpacingValue Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : ModifyD1SpacingValue Method (IInstanceToVaryOptions)

Modifies the spacing in Direction 1 of the specified pattern instance.

.NET Syntax

Visual Basic (Declaration)

```
Function ModifyD1SpacingValue( _
    ByVal Instance As System.Integer, _
    ByVal Value As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions
Dim Instance As System.Integer
Dim Value As System.Double
Dim value As System.Boolean
```

```
value = instance.ModifyD1SpacingValue(Instance, Value)
```

C#

```
System.bool ModifyD1SpacingValue(
    System.int Instance,
    System.double Value
)
```

C++/CLI

```
System.bool ModifyD1SpacingValue(
    & System.int Instance,
    & System.double Value
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

Value

Spacing value in Direction 1

Return Value

True if the spacing is modified successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::ModifyD1SpacingValue](#).

Example

See the [IInstanceToVaryOptions](#) examples.

Remarks

To calculate Instance:

1. Ensure that the D2PatternSeedOnly property in the pattern feature data object is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (j - 1) + (i - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- *i* = pattern instance number
- *n2* = number of instances in Direction 2
- *i* = index for Direction 1
- *j* = index for Direction 2

In the pattern's PropertyManager, find *n2* in the **Direction 2 > Spacing and Instances > Number of instances** field and find [*i,j*] in the **Modified Instances** section.

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)

Availability

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SOLIDWORKS API Help

[Send comments](#) on this topic.

ModifyD2SpacingValue Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : ModifyD2SpacingValue Method (IInstanceToVaryOptions)

Modifies the spacing in Direction 2 of the specified pattern instance.

.NET Syntax

Visual Basic (Declaration)

```
Function ModifyD2SpacingValue( _  
    ByVal Instance As System.Integer, _  
    ByVal Value As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions  
Dim Instance As System.Integer  
Dim Value As System.Double  
Dim value As System.Boolean
```

```
value = instance.ModifyD2SpacingValue(Instance, Value)
```

C#

```
System.bool ModifyD2SpacingValue(  
    System.int Instance,  
    System.double Value  
)
```

C++/CLI

```
System.bool ModifyD2SpacingValue(  
& System.int Instance,  
& System.double Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

Value

Spacing value in Direction 2

Return Value

True if the spacing is modified successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [IInstanceToVaryOptions::ModifyD2SpacingValue](#).

Example

See the [IInstanceToVaryOptions](#) examples.

Remarks

To calculate Instance:

1. Ensure that the D2PatternSeedOnly property in the pattern feature data object is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (i - 1) + (j - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- I = pattern instance number
- $n2$ = number of instances in Direction 2
- i = index for Direction 1
- j = index for Direction 2

In the pattern's PropertyManager, find $n2$ in the **Direction 2 > Spacing and Instances > Number of instances** field and find $[i,j]$ in the **Modified Instances** section.

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)

Availability

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[Send comments on this topic.](#)

ModifyDimensions Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : ModifyDimensions Method (IInstanceToVaryOptions)

Modifies the specified dimensions of the specified pattern instance.

.NET Syntax

Visual Basic (Declaration)

```
Function ModifyDimensions( _  
    ByVal Instance As System.Integer, _  
    ByVal Dimensions As System.Object, _  
    ByVal Values As System.Object _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions  
Dim Instance As System.Integer  
Dim Dimensions As System.Object  
Dim Values As System.Object  
Dim value As System.Boolean
```

```
value = instance.ModifyDimensions(Instance, Dimensions, Values)
```

C#

```
System.bool ModifyDimensions(  
    System.int Instance,  
    System.object Dimensions,  
    System.object Values  
)
```

```
System.bool ModifyDimensions(
& System.int Instance,
& System.Object^ Dimensions,
& System.Object^ Values
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Instance

Pattern instance number (see **Remarks**)

Dimensions

Array of dimensions

Values

Array of values

Return Value

True if dimensions modified successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::ModifyDimensions](#).

Example

See the [InstanceToVaryOptions](#) examples.

Remarks

To calculate *Instance*:

1. Ensure that the D2PatternSeedOnly property in the pattern feature data object is set to false.
2. Calculate:

For a bi-directional pattern:

$$I = n2 * (j - 1) + (i - 1)$$

For a uni-directional pattern:

$$I = i - 1$$

where:

- *I* = pattern instance number
- *n2* = number of instances in Direction 2
- *i* = index for Direction 1
- *j* = index for Direction 2

In the pattern's PropertyManager, find *n2* in the **Direction 2 > Spacing and Instances > Number of instances** field and find [*i,j*] in the **Modified Instances** section.

See Also

[IInstanceToVaryOptions Interface](#)

[IInstanceToVaryOptions Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

SetD1IncrementDimensions Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : SetD1IncrementDimensions Method (IInstanceToVaryOptions)

Sets the dimensions to increment in Direction 1.

.NET Syntax

Visual Basic (Declaration)

```
Function SetD1IncrementDimensions( _
    ByVal IncrementDimensions As System.Object, _
    ByVal IncrementDimValues As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions
Dim IncrementDimensions As System.Object
Dim IncrementDimValues As System.Object
Dim value As System.Boolean
```

```
value = instance.SetD1IncrementDimensions(IncrementDimensions, IncrementDimValues)
```

C#

```
System.bool SetD1IncrementDimensions(
    System.object IncrementDimensions,
    System.object IncrementDimValues
)
```

C++/CLI

```
System.bool SetD1IncrementDimensions(
    & System.Object^ IncrementDimensions,
    & System.Object^ IncrementDimValues
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncrementDimensions
Array of dimensions to increment
IncrementDimValues
Array of dimension increments in Direction 1

Return Value

True if the dimensions are incremented successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::SetD1IncrementDimensions](#).

Example

See the [IInstanceToVaryOptions](#) examples.

See Also

[IInstanceToVaryOptions Interface](#)
[IInstanceToVaryOptions Members](#)

Availability

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SOLIDWORKS API Help

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SetD2IncrementDimensions Method (IInstanceToVaryOptions)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInstanceToVaryOptions Interface](#) : SetD2IncrementDimensions Method (IInstanceToVaryOptions)

Sets the dimensions to increment in Direction 2.

.NET Syntax

Visual Basic (Declaration)

```
Function SetD2IncrementDimensions( _
    ByVal IncrementDimensions As System.Object, _
    ByVal IncrementDimValues As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInstanceToVaryOptions
Dim IncrementDimensions As System.Object
Dim IncrementDimValues As System.Object
Dim value As System.Boolean
```

```
value = instance.SetD2IncrementDimensions(IncrementDimensions, IncrementDimValues)
```

C#

```
System.bool SetD2IncrementDimensions(
    System.object IncrementDimensions,
    System.object IncrementDimValues
)
```

C++/CLI

```
System.bool SetD2IncrementDimensions(
    & System.Object^ IncrementDimensions,
    & System.Object^ IncrementDimValues
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IncrementDimensions

Array of dimensions to increment

IncrementDimValues

Array of dimension increments in Direction 2

Return Value

True if the dimensions are incremented successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [InstanceToVaryOptions::SetD2IncrementDimensions](#).

See Also

[IInstanceToVaryOptions Interface](#)

[InstanceToVaryOptions Members](#)

Availability

SOLIDWORKS 2021 FCS, Revision Number 29

IInterference (Type)

IInterference Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IInterference Interface

Allows access to the components that interfere when interference detection is calculated.
NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IInterference
```

Visual Basic (Usage)



```
Dim instance As IInterference
```

C#

```
public interface IInterference
```

C++/CLI

```
public interface class IInterference
```

Visual Basic for Applications (VBA) Syntax

See [Interference](#).

Example

[Run Interference Detection \(C#\)](#)
[Run Interference Detection \(VB.NET\)](#)
[Run Interference Detection \(VBA\)](#)

Accessors

[IInterferenceDetectionMgr::GetInterferences](#) and [IInterferenceDetectionMgr::IGetInterferences](#)

Access Diagram

[Interference](#)

See Also

[IInterference Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IInterferenceDetectionMgr Interface](#)
[ICollision Interface](#)

IInterference Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IInterference Interface

The following tables list the members exposed by [Interference](#).

Public Properties

Name	Description
 Components	Gets the components that interfere with each other.
Ignore	Gets or sets whether to ignore this interference.

[IsFastener](#)

Gets whether this interference includes a fastener or not.

[IsPossibleInterference](#)

Gets whether this interference is a possible interference.

[Volume](#)

Gets the volume of the interference.

[Top](#)

Public Methods

Name	Description
 GetComponentCount	Gets the number of components interfering with each other.
 GetInterferenceBody	Gets the interfering volume.
 IGetComponents	Gets the components that interfere with each other.

[Top](#)

See Also

[IInterference Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[InterferenceDetectionMgr Interface](#)
[ICollision Interface](#)

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IInterference Interface Methods

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) : IInterference InterfaceFor a list of all members of this type, see [IInterference members](#).

Public Methods

Name	Description
 GetComponentCount	Gets the number of components interfering with each other.
 GetInterferenceBody	Gets the interfering volume.
 IGetComponents	Gets the components that interfere with each other.

[Top](#)

See Also

[IInterference Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[InterferenceDetectionMgr Interface](#)
[ICollision Interface](#)

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IInterference Interface Properties

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) : IInterference InterfaceFor a list of all members of this type, see [IInterference members](#).

Public Properties

Name	Description
 Components	Gets the components that interfere with each other.
 Ignore	Gets or sets whether to ignore this interference.
 IsFastener	Gets whether this interference includes a fastener or not.
 IsPossibleInterference	Gets whether this interference is a possible interference.
 Volume	Gets the volume of the interference.

[Top](#)

See Also

[IInterference Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[InterferenceDetectionMgr Interface](#)
[ICollision Interface](#)

SOLIDWORKS API Help

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Components Property (IInterference)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : Components Property (IInterference)

Gets the components that interfere with each other.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Components As System.Object

Visual Basic (Usage)



```
Dim instance As IInterference
Dim value As System.Object

value = instance.Components
```

C#

```
System.Object Components {get;}
```

C++/CLI

```
property System.Object^ Components {
    System.Object^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of the [components](#) that interfere with each other

Visual Basic for Applications (VBA) Syntax

See [Interference::Components](#).

Example

[Run Interference Detection \(C#\)](#)
[Run Interference Detection \(VB.NET\)](#)
[Run Interference Detection \(VBA\)](#)

See Also

[IInterference Interface](#)
[IInterference Members](#)
[Interference::GetComponentCount Method](#)
[Interference::IGetComponents Method](#)
[InterferenceDetectionMgr::GetInterferenceComponentCount Method](#)
[InterferenceDetectionMgr::GetInterferenceComponents Method](#)
[InterferenceDetectionMgr::IGetInterferenceComponents Method](#)
[InterferenceDetectionMgr::MakeInterferingPartsTransparent Property](#)
[InterferenceDetectionMgr::NonInterferingComponentDisplay Property](#)
[InterferenceDetectionMgr::TreatSubAssembliesAsComponents Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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GetComponentCount Method (IInterference)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : GetComponentCount Method (IInterference)

Gets the number of components interfering with each other.

.NET Syntax

Visual Basic (Declaration)

```
Function GetComponentCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IInterference  
Dim value As System.Integer  
  
value = instance.GetComponentCount()
```

C#

```
System.int GetComponentCount()
```

C++/CLI

```
System.int GetComponentCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of components interfering with each other

Visual Basic for Applications (VBA) Syntax

See [Interference::GetComponentCount](#).

Example

[Run Interference Detection \(C#\)](#)
[Run Interference Detection \(VB.NET\)](#)
[Run Interference Detection \(VBA\)](#)

Remarks

Call this method before calling [IInterference::IGetComponents](#).

See Also

[IInterference Interface](#)
[IInterference Members](#)
[Interference::Components Property](#)
[InterferenceDetectionMgr::GetInterferenceComponentCount Method](#)
[InterferenceDetectionMgr::GetInterferenceComponents Method](#)
[InterferenceDetectionMgr::IGetInterferenceComponents Method](#)
[InterferenceDetectionMgr::MakeInterferingPartsTransparent Property](#)
[InterferenceDetectionMgr::NonInterferingComponentDisplay Property](#)
[InterferenceDetectionMgr::TreatSubAssembliesAsComponents Property](#)

Availability

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SOLIDWORKS API Help

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GetInterferenceBody Method (IInterference)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : GetInterferenceBody Method (IInterference)

Gets the interfering volume.

.NET Syntax

Visual Basic (Declaration)

```
Function GetInterferenceBody() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IInterference  
Dim value As System.Object  
  
value = instance.GetInterferenceBody()
```

C#

```
System.object GetInterferenceBody()
```

C++/CLI

```
System.Object^ GetInterferenceBody();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Interfering volume](#)

Visual Basic for Applications (VBA) Syntax

See [Interference::GetInterferenceBody](#).

See Also

[IInterference Interface](#)
[Interference Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetComponents Method (IInterference)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : IGetComponents Method (IInterference)

Gets the components that interfere with each other.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetComponents( _  
    ByVal ComponentCount As System.Integer _  
) As Component2
```

Visual Basic (Usage)



```
Dim instance As IInterference  
Dim ComponentCount As System.Integer  
Dim value As Component2  
  
value = instance.IGetComponents(ComponentCount)
```

C#

```
Component2 IGetComponents(  
    System.int ComponentCount  
)
```

C++/CLI

```
Component2^ IGetComponents(  
& System.int ComponentCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ComponentCount
Number of components interfering with each other

Return Value

- in-process, unmanaged C++: Pointer to an array of [components](#) interfering with each other
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [IInterference::GetComponentCount](#) before calling this method to determine the size of the array.

See Also

[IInterference Interface](#)
[IInterference Members](#)
[IInterference::Components Property](#)
[IInterferenceDetectionMgr::GetInterferenceComponentCount Method](#)
[IInterferenceDetectionMgr::GetInterferenceComponents Method](#)
[IInterferenceDetectionMgr::IGetInterferenceComponents Method](#)
[IInterferenceDetectionMgr::MakeInterferingPartsTransparent Property](#)
[IInterferenceDetectionMgr::NonInterferingComponentDisplay Property](#)
[IInterferenceDetectionMgr::TreatSubAssembliesAsComponents Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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Ignore Property (IInterference)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : Ignore Property (IInterference)

Gets or sets whether to ignore this interference.

.NET Syntax

Visual Basic (Declaration)

```
Property Ignore As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInterference  
Dim value As System.Boolean
```

```
instance.Ignore = value
```

```
value = instance.Ignore
```

C#

```
System.bool Ignore {get; set;}
```

C++/CLI

```
property System.bool Ignore {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to ignore this interference, false to not

Visual Basic for Applications (VBA) Syntax

See [Interference::Ignore](#).

See Also

[IInterference Interface](#)
[IInterference Members](#)
[InterferenceDetectionMgr::ShowIgnoredInterferences Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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IsFastener Property (IInterference)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : IsFastener Property (IInterference)

Gets whether this interference includes a fastener or not.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property IsFastener As System.Boolean

Visual Basic (Usage)



```
Dim instance As IInterference
Dim value As System.Boolean

value = instance.IsFastener
```

C#

```
System.bool IsFastener {get;}
```

C++/CLI

```
property System.bool IsFastener {
    System.bool get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the interference includes a fastener, false if not

Visual Basic for Applications (VBA) Syntax

See [Interference::IsFastener](#).

See Also

[IInterference Interface](#)
[IInterference Members](#)
[InterferenceDetectionMgr::CreateFastenersFolder Property](#)

Availability

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SOLIDWORKS API Help

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IsPossibleInterference Property (IInterference)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : IsPossibleInterference Property (IInterference)

Gets whether this interference is a possible interference.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property IsPossibleInterference As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IInterference  
Dim value As System.Boolean
```

```
value = instance.IsPossibleInterference
```

C#

```
System.bool IsPossibleInterference {get;}
```

C++/CLI

```
property System.bool IsPossibleInterference {  
    System.bool get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if this is a possible interference, false if not

Visual Basic for Applications (VBA) Syntax

See [Interference::IsPossibleInterference](#).

See Also

[IInterference Interface](#)
[IInterference Members](#)

Availability

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Volume Property (IInterference)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IInterference Interface](#) : Volume Property (IInterference)

Gets the volume of the interference.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Volume As System.Double
```

Visual Basic (Usage)



```
Dim instance As IInterference  
Dim value As System.Double
```

```
value = instance.Volume
```

C#

```
System.double Volume {get;}
```

C++/CLI

```
property System.double Volume {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Volume of the interference

Visual Basic for Applications (VBA) Syntax

See [Interference::Volume](#).

Example

[Run Interference Detection \(C#\)](#)
[Run Interference Detection \(VB.NET\)](#)
[Run Interference Detection \(VBA\)](#)

See Also

[Interference Interface](#)
[Interference Members](#)

Availability

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